

## Summer23: CSE320

### Section 11

#### Quiz 1

18 June, 2023

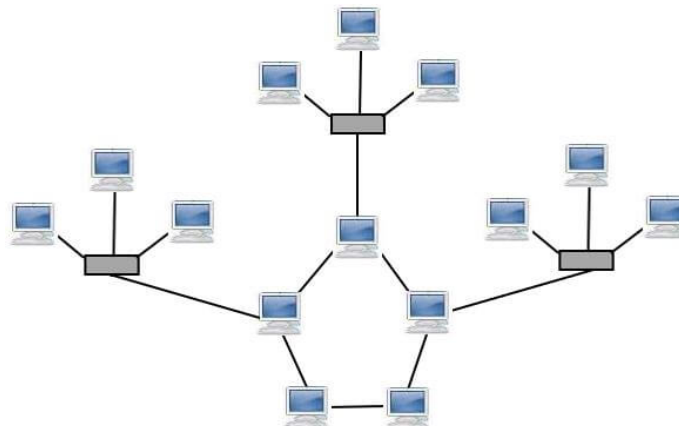
Total Marks: 20

**Student ID:**

**Name:**

1. Put a tick (✓) beside the correct answer. (5 Marks)
  - a. What is the number of bits in a PORT address?
    - i) 8
    - ii) 16
    - iii) 32
    - iv) 48
  - b. Which of the following is a valid IP address?
    - i) 103.256.045.78
    - ii) 254.255.200.193
    - iii) 03.03.03.03
    - iv) 3.3.3.3.3
  - c. What is the number of bits in an IP address?
    - i) 16
    - ii) 32
    - iii) 48
    - iv) 64
  - d. You're using your lab computer connected to the BRACU network. Which of the following network types best describe this network?
    - i) PAN
    - ii) MAN
    - iii) LAN
    - iv) WAN
  - e. If there are 5 nodes connected in a hub topological network, what is the required number of links between the nodes?
    - i) 4
    - ii) 5
    - iii) 10
    - iv) 15

2. Identify all the topologies used in the following network. Mention the topology names and briefly explain the reasons behind your answer. (4 Marks)



3. What are the differences between OSI model and TCP/IP model? Explain in brief. (4 Marks)

4. In the following network, Letters denote IP addresses and Numbers denote MAC addresses. Write down the route of a frame sent from the sender to the receiver and demonstrate how MAC address changes by showing frame header contents for each hop. Frame header will contain Destination MAC, Source MAC, Destination IP, Source IP respectively. (7 Marks)

